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RESEARCH ARTICLE

Assessing Barriers to the Adoption of Green Technologies in Developing Countries by Using an Integrated Decision-Making System

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Received: 22 October 2025 | **Revised:** 29 November 2025 | **Accepted:** 7 December 2025

Keywords: cleaner production | climate change | decision making | green investment | green technologies barriers | sustainable development goals

ABSTRACT

The United Nations Sustainable Development Goals (SDGs) set a global agenda to advance sustainability by exerting environmental pressures and promoting cleaner production. Achieving these targets depends heavily on the adoption of green technologies, yet several persistent barriers continue to impede the effective implementation of green technologies. Therefore, this study develops an integrated decision support system to assess Green Technology Adoption (GTA) barriers and explore potential strategies to overcome them effectively. After reviewing the literature and consulting with field experts, seven major categories of barriers: institutional, financial, market, technological, regulatory, environmental, and social barriers, and 26 sub-barriers were identified and analyzed. A Fuzzy Analytic Hierarchy Process (FAHP) was applied to rank the main and sub barriers and then used the Fuzzy VIekriterijumsko KOmpromisno Rangiranje (FVIKOR) method to prioritize the 11 potential solution strategies. The results reveal that financial barriers (FB), technological barriers (TB), and regulatory barriers (RB) are the most pressing, whereas natural resource dependency (EB1), limited institutional capacity (IB3), and climate challenges (EB2) were ranked the most influencing sub-barriers. Moreover, resilience programs for green tech jobs (SO7) and investment in digital technologies (SO6) ranked the top potential strategic solutions to mitigate the GTA barriers. This study provides actionable guidance for developing economies pursuing green transitions under economic and governance constraints, offering policymakers and practitioners a clear, systematic roadmap for designing resilient sustainability strategies.

1 | Introduction

Over the past few years, the growing urgency of climate change and environmental degradation has made the adoption of green technologies not just desirable but indispensable. Achieving the United Nations Sustainable Development Goals (SDGs) largely depends on how effectively countries are able to integrate such innovations into their development strategies (Ikram et al. 2021). According to the International Energy Agency (IEA 2023), improving the efficiency of infrastructure, promoting cleaner transportation, and greening energy systems can substantially reduce greenhouse gas emissions, making it an important step in the global fight against climate change.

Beyond their environmental benefits, green technologies also generate positive social and economic outcomes, which explains the reason why they are increasingly seen as pillars of sustainable development. As global temperatures continue to rise, many nations have intensified their own initiatives to confront the climate crisis, leading to accelerated progress in renewable energy innovation (Belaïd et al. 2023). The growing reliance on renewable electricity not only lessens dependence on finite natural resources but also helps maintain ecological balance and supports the ongoing transition toward a low-carbon global economy. These enhancements are necessary in order to reduce greenhouse gas emissions, ease energy shortages, and address broader issues of environmental deterioration (Zhao, Gao, et al. 2024).

The sharp rise in carbon dioxide (CO_2) emissions, which is largely driven by the expansion of non-renewable energy sources such as fossil fuels, has become one of the most pressing global challenges of our time. The world's continued dependence on fossil fuels has not only fueled rapid growth in energy consumption but has also intensified CO_2 emissions, heightening environmental concerns (Miao et al. 2025). This trend contributes directly to global warming and is amplified by broader social and economic shifts, including industrial modernization, population growth, urbanization, and changing consumption behaviors.

In response, there is growing momentum toward the development and adoption of cleaner, renewable energy technologies. This transition is no longer a choice but a necessity to be able to reduce carbon emissions and align national policies with the principles of sustainable development. As Moreover, advancing environmentally responsible technologies is central to building a more climate-resilient and sustainable global future (Sajid et al. 2024).

Adopting green technologies represents a paradigm change which integrates economic success with environmental stewardship and attracts substantial interest from academics and is increasingly being integrated into conventional business practices via specialized frameworks across various nations. This transition fundamentally reshapes economic and social paradigms: economies move from carbon-intensive industries to sustainable, innovation-driven models, production processes become circular and resource-efficient, and societal values increasingly prioritize long-term environmental well-being (Suki et al. 2022). Green technologies are mainly characterized as molding the world, facilitating economic advancement while maintaining environmental accountability (Zhou et al. 2015). By dissociating development from carbon-intensive practices, these innovations directly contribute to several SDGs specifically SDG 7 (Affordable and Clean Energy), SDG 11 (Sustainable Cities and Communities), SDG 12 (Responsible Consumption and Production), and SDG 13 (Climate Action) (Anzolin and Lebdioui 2021). Adopting renewable energy and energy-efficient technologies directly supports SDG 7 by expanding access to clean power and improving energy efficiency in developing regions (Estevão and Lopes 2024). Green infrastructure and low-emission transport contribute to SDG 11 by reducing pollution and enhancing urban sustainability (Patel et al. 2024). Likewise, cleaner production methods and recycling technologies promote SDG 12 through improved resource efficiency, whereas widespread deployment of emissions-reducing innovations is crucial for SDG 13 (Wang, Osei, and Agyemang 2025; Wang, Pal, et al. 2025; Wang, Li, et al. 2025).

The development and implementation of climate change adaptation and mitigation technologies is also highlighted in international assessments carried out by international organizations such as the United Nations Environment Program (UNESCAP 2020). Such evaluations are being considered as essential to the response to the rapidly escalating problem of climate change. A socioeconomic potential for fostering sustainable development on a worldwide scale is presented by the transition to environmentally friendly technology, which is not only a necessity for the environment but also a necessity for the economy. To accomplish the objectives of global sustainability, it is essential to identify the barriers which are inhibiting the adoption of environmentally friendly technologies and to strategically overcome them. The transition toward environmentally responsible innovation is typically hampered by structural and behavioral limits that influence investment, policy, and societal action (Maghyereh et al. 2025). These restrictions, in addition to the technical and financial factors, are significant barriers. There are two primary categories of such barriers: psychological and functional. Functional constraints include factors like poor infrastructure, inconsistent regulations, inadequate finance schemes, and limited technology. These elements determine an economy's structural capacity to implement green transformations. On the other hand, psychological barriers are related to society's attitudes, beliefs, and incentives that affect how responsive institutions are, how consumers act, and how committed politicians are. These indicators imply that the success of global green technologies deployment depends on both physical assets and governance structures, as well as a social and cultural readiness to accept sustainable change (Nketiah et al. 2024). Acknowledging and tackling these different aspects leads to a more thorough comprehension of the motives behind the global green transition's frequent inability to meet set sustainability objectives and underscores the necessity of combined policy, education, and innovation initiatives for sustainable development.

Across developing economies, energy transitions proceed under tighter fiscal space, higher capital costs, and uneven institutional capacity. Governments are pursuing mixed pathways that pair rapid scale-up of renewables with interim reliability measures such as grid reinforcement and flexible generation, whereas catalyzing industrial decarbonization through standards, concessional finance, and public-private partnerships (Olaniyi et al. 2025). Anchor firms and state-owned utilities often act as early movers, de-risking technologies and crowding in private investment, yet diffusion is frequently slowed by tariff design, currency risk, technology access, and skills gaps. These realities underscore a central implementation challenge: translating national ambition into bankable projects and routine practice in settings where functional readiness infrastructure, regulation, finance, and technology, and motivational readiness incentives, norms, and institutional responsiveness vary widely. The maturity of green technology adoption is closely tied to the presence of digital ecosystems and platform-based innovation, which enhance the scalability and systemic integration of green solutions (Sassanelli et al. 2022). Recognizing this heterogeneity, we adopt a comparative lens on developing economies to identify which barriers most constrain adoption and which interventions yield the greatest marginal gains under real-world constraints.

The transition to green technologies still faces many significant obstacles, especially in developing nations regardless of the ongoing global initiatives (Hernández Soto 2024). Financial limitations, technological disparities, institutional deficiencies, and socio-cultural aversion usually hinder the widespread adoption and successful execution of green technologies; by recognizing these obstacles and identifying appropriate solutions, their implementation can promote an inclusive, efficient, and prompt green transformation. The purpose of this study is to assess green technology barriers and recommend practical solutions.

The study is guided by the research question: Which barriers most critically impede Green Technology Adoption (GTA) in a developing country, and what integrated strategies can effectively overcome these barriers? In addressing this question, we make two key contributions. First, we develop an integrated GTA strategic framework and analyze seven major categories and twenty-six sub-barriers. First, we develop an integrated GTA strategic framework and analyzed seven major categories and 26 sub-barriers by Fuzzy Analytic Hierarchy Process (FAHP) was applied to rank the main and sub-barriers and, then used the Fuzzy VIekriterijumsko KOmpromisno Rangiranje (FVIKOR) methods. This hybrid approach leverages FAHP to derive rigorous weights for each barrier category and FVIKOR to rank potential solutions, thereby combining the strengths of hierarchical weighting and compromise ranking in one decision-making system. This integration offers more robust and balanced insights. Second, through an empirical application, this study provides actionable policy recommendations and a prioritized roadmap that can inform stakeholders in other developing economies. These contributions advance the literature on GTA by bridging methodological gaps and delivering practical, context-sensitive findings.

The study is organized as follows: Section 2 presents the comprehensive literature review of main barriers and sub-barriers of GTA. The FAHP and FVIKOR construction steps are detailed in Section 3. The findings and analysis are explained in Section 4. Section 5 outlines the discussion and policy implications, whereas the conclusion, along with limitations and future directions, are presented in the last section of the paper.

2 | Literature Review

The growing challenges of climate change, environmental degradation, and resource scarcity have placed green technologies at the center of global sustainability discussions. These technologies are often described as green innovation, eco-innovation, or environmental innovation, referring to products, processes, and systems designed to reduce environmental harm while supporting economic and social progress. Although the importance of these concepts is widely recognized, their definitions remain somewhat inconsistent in academic and policy literature, leading to different interpretations and areas of emphasis among scholars and institutions (Omisore 2018; Wang and Azam 2024). In this study, green technology is defined as any technology or innovation designed to mitigate environmental impacts and promote sustainability, such as renewable energy systems, pollution control, resource-efficient processes. This definition aligns with recent literature and encompasses concepts sometimes labeled as eco-innovation or green innovation, which we treat equivalently in scope (Oduro et al. 2022).

Green technologies can be understood as a broad range of technical solutions such as renewable energy systems, energy-efficient infrastructure, clean production methods, and pollution-control technologies that directly contribute to the global environmental objectives. At the same time, they represent forms of innovation that go beyond technology itself, extending to organizational practices, public policy design, and even individual behavior. Because of this overlap, many scholars describe the terms green innovation, eco-innovation, and green technology as closely connected, sometimes even used interchangeably within the literature (Faucheux and Nicolaï 2011; Jiang et al. 2021).

Building upon this conceptual framework, recent research has demonstrated that corporate actors are increasingly promoting the adoption and development of green technologies by employing targeted green innovation strategies. Scholars are increasingly emphasizing that the eventual direction of corporate sustainability management will require the establishment of sustainable development practices, carbon emission reduction, and green transformation (Zhao et al. 2025). In this regard, corporate green innovation plays a key role in improving the environmental performance of firms. This form of innovation involves not only the development of cleaner technologies but also notable progress in information systems, digital platforms, and e-commerce activities that help companies adapt to growing global competition. These elements highlight the strategic importance of embedding green innovation within core business operations, allowing sustainability to function both as an environmental necessity and as a potential source of long-term competitive advantage (Le et al. 2024). Particularly, green technologies have the potential to revolutionize the energy sector by lowering energy demand, improving efficiency, and promoting the development of cost-effective renewable projects. Policymakers worldwide are recognizing the critical role of green technologies in promoting the adoption of renewable energy and optimizing resource use and are implementing forward-thinking strategies to incorporate environmentally sustainable innovations into the energy sector (Behera et al. 2024).

The ongoing shift toward green technologies is not without challenges, as traditional industries often resist transformation, particularly when new innovations begin to gain economic relevance. In addition, variations in the availability of natural resources can negatively influence investments in renewable energy, sometimes even more strongly than general economic conditions. Despite these barriers, the expansion of clean energy continues to be supported by steady technological progress, favorable market dynamics, and improvements in energy efficiency. Although volatility in the financial markets appears to have a limited direct impact, the adoption of renewable energy has been strongly encouraged by advancements in ecological innovation (Han et al. 2025). The relationship between economic development, foreign direct investment (FDI), and green technology remains intricate and context-dependent (Hernández Soto 2024). Although it may occasionally impede the adoption of renewable energy in the short-term, it can also contribute to the long-term reduction of carbon emissions and energy consumption. Environmental and economic benefits are further reinforced by policy mechanisms, such as revenue-neutral taxation that concentrates on fossil fuel-based energy generation. All together, these dynamics emphasize the essential role of integrated innovation and policy in facilitating sustainable energy transitions in developing nations (Radtke 2025).

Moreover, it is important to note that the critical role of Small and Medium-sized Enterprises (SMEs) in advancing green technologies within developing economies is significant. SMEs account for the majority of businesses and thus have a substantial cumulative environmental impact, yet they face unique barriers to green innovation. Recent studies indicate that SMEs often struggle with limited financial resources and technical expertise, as well as inadequate institutional support, which hinder their ability to implement sustainable technologies (Yacob et al. 2019).

Further green technologies not only support a shift toward low-carbon energy systems but also optimize the use of sustainable resources (Lv et al. 2024). The focal point of green innovation is the development and implementation of technologies that minimize the environmental impact at every stage, from production to use; these advancements are intended to not only mitigate damage but also to actively monitor and resolve it at the source. The structured use of design rules within product-service systems facilitates coordinated knowledge and lifecycle integration principles that directly support green technology adoption by enabling more sustainable design and operational practices across business units (Sassanelli et al. 2018).

Green technologies play a vital role in advancing sustainability by improving energy efficiency, cutting emissions, and enabling better management of environmental impacts. However, many of these goals remain difficult to achieve through traditional, non-green technologies alone. For this reason, accelerating progress in green innovation has become essential to make sustainable development genuinely attainable and to move closer to a model of growth that is both economically viable and environmentally responsible (Islam 2025).

Recent research highlights several major barriers that make it difficult to adopt green technologies on a large scale. Prior studies highlight five major challenges in advancing the green transition. First, real transformation requires radical innovation rather than small, incremental improvements (Nabi et al. 2023). Second, uncertainty in markets remains high, as many sectors still operate under a business-as-usual mindset, which makes investment in green technologies both risky and unpredictable (Roper and Tapinos 2016). Third, it is not easy for governments to design policies that are consistent, long-term, and effective in promoting green innovation (Wang et al. 2022). Fourth, the transition also raises questions of social fairness, since the costs and benefits of environmental actions are not equally distributed across different groups (Ternes et al. 2024). Lastly, environmental risks are global by nature, which means that no single country or company can address them alone (Benzaken et al. 2024). Together, these interconnected challenges suggest that moving toward a green economy will require not only new technologies but also deeper structural and institutional changes. Recent systematic reviews further confirm that financial constraints, technological limitations, policy uncertainty, and weak institutional support remain among the main barriers to GTA globally and highlighted the importance of culturally sensitive approaches, enhanced community engagement, and access to financing and technical expertise to overcome similar obstacles (Al-Emran and Griffy-Brown 2023; Zheng et al. 2024). The detailed summary of GTA barriers and sub-barriers is presented in Table 1. The description of each GTA barrier is presented in (see Supporting Information: Table A2).

This study developed an integrated framework that represents a holistic, multi-dimensional model for GTA for developing economies (see Figure 1). It positions GTA as the central outcome, driven by the integration of several readiness dimensions that address the spectrum of barriers typically impeding sustainability innovations in these regions. Specifically, the framework integrates functional readiness encompassing technical competence, infrastructure, and operational capability, motivational readiness, the organizational commitment and drive to adopt green innovations, compatibility, the fit of new technology with existing processes and values, and synergy across business units, cross-departmental collaboration and knowledge sharing. These internal dimensions are conceptually supported by an integration layer that binds them into a cohesive strategy, and the entire system is encircled by external environmental and stakeholder monitoring factors that continually inform and influence the adoption process. This integrated design is a novel theoretical contribution, directly responding to calls in the literature for more systemic approaches; prior studies have noted that green transitions involve multiple factors, and it is not enough to analyze only from one perspective or through one factor (Haaja and Harikkala-Laihinen 2025). By unifying technical, organizational, and contextual elements, the framework systematically tackles key barriers: financial, regulatory, technological, institutional, environmental, and social that have been widely documented in developing economies. In contrast to fragmented models, this comprehensive framework offers a central guiding theory for understanding and overcoming the complex, interrelated obstacles to green technology uptake in less developed settings. Beyond core readiness, the framework introduces compatibility and synergy across business units as key dimensions that ensure the green technology can be embedded effectively within the organization. Compatibility denotes the degree to which a new green technology aligns with the firm's existing processes, values, and needs. High compatibility smooths the adoption process, as innovations perceived to be consistent with current practices face less resistance (Xia et al. 2019). In the context of sustainable technology, compatibility is crucial because many green practices are additions to or modifications of a company's established operations. A technology that fits well with local circumstances and organizational culture helps overcome social and behavioral barriers such as user resistance or misalignment with norms. Meanwhile, synergy across business units emphasizes breaking organizational silos so that different departments or units work collaboratively in the adoption effort. Developing economies often struggle with siloed implementation of new technologies, which can lead to fragmented outcomes and lost opportunities for scale (Sethi et al. 2024). The framework's inclusion of cross-unit synergy addresses this by promoting internal collaboration and knowledge sharing. Literature on green innovation management has highlighted that cross-functional integration within the firm is a critical enabler for sustainability-oriented innovations, as it facilitates the flow of environmental knowledge and coordinated action across various divisions (Ul-Durar et al. 2023; Sassanelli et al. 2021). When R&D, operations, marketing, and other units are jointly engaged, firms can leverage diverse expertise and ensure that the green technology is compatible across the entire enterprise.

TABLE 1 | Summary of green technology barriers and sub-barriers.

Main barrier	Code	Sub-barriers	Code	Source
Financial barriers	FB	Lack of access to financial resources	FB 1	Solangi et al. (2021), Sardianou and Kostakis (2020), and Mirza et al. (2024)
		High-cost premium	FB 2	
		Unattractive returns on investment (ROI)	FB 3	
		Unavailability of government subsidies	FB 4	
		High operational and maintenance costs	FB 5	
Regulatory barriers	RB	Unclear regulations	RB1	Painuly and Wohlgemuth (2020) and Wang, Li, and Liu (2023)
		Lack of government support/incentives	RB2	
		Ineffective enforcement of environmental laws	RB3	
		Complex and rigid legislation	RB4	
Technological barriers	TB	Lack of local expertise and R&D	TB 1	Oryani et al. (2021), Jafarizadeh et al. (2024) and Akusta et al. (2025)
		Obsolete infrastructure	TB 2	
		Limited access to green technologies	TB 3	
Market barriers	MB	Low demand for sustainable products	MB 1	Du et al. (2025), Bang-Ning et al. (2025), and Barbieri et al. (2025)
		Limited awareness of green technologies	MB 2	
		Market resistance to change	MB 3	
		Unequal distribution of benefits	MB 4	
Institutional barriers	IB	Bureaucratic inefficiency	IB 1	Jiang et al. (2021), Yang et al. (2022), and Healey et al. (2017)
		Poor coordination	IB 2	
		Limited institutional capacity	IB 3	
		Political instability	IB 4	
Environmental barriers	EB	Natural resource dependency	EB 1	Tantawat et al. (2023), Gao et al. (2024), and Guo, He, et al. (2024)
		Climate challenges	EB 2	
		Limited natural resource availability	EB 3	
Social barriers	SB	Public resistance to change	SB 1	Jang et al. (2024), de Soysa (2025), and Guo, Dong, et al. (2024)
		Low stakeholder engagement	SB 2	
		Social inequalities	SB 3	

This integrated approach mitigates barriers related to organizational structure and culture. Moreover, it reduces internal resistance to change by fostering a shared vision. The compatibility and synergy dimensions of the framework ensure that once a firm is functionally and motivationally ready, the innovation is organizationally feasible and supported throughout the business. This integrated attention to internal fit and cooperation is part of the framework's unique contribution, as it systematically engineers an environment where green innovations can thrive rather than flounder due to internal frictions. The proposed integrated framework offers a systemic, novel approach: it develops internal capacity-building and motivation with organizational alignment and external vigilance. In doing so, it provides a theoretical foundation for overcoming the multifaceted barriers to GTA in developing economies, emphasizing that true progress requires a synchronized effort across all levels of the firm and its environment. Further, most existing models of GTA either focus on identifying and ranking barriers or on evaluating solutions, but seldom both together. Our proposed integrated framework extends these models by unifying the two stages: It first quantifies barrier importance under uncertainty via fuzzy AHP and then directly feeds those insights into a fuzzy VIKOR analysis to rank solutions. This dual-stage integration challenges the traditional siloed approach and provides a more holistic tool for decision-makers, capturing the interplay between barrier severity and solution efficacy within one model.

3 | Research Method

The GTA barriers were chosen in light of developing economic structure, governmental regulations, and the necessity of implementing green technologies to promote sustainable development. This study identified 26 sub-barriers under seven main categories of GTA barriers. We adopted this categorization framework based on recurring themes in the literature. Thus, the basis of our barrier classification is a combination

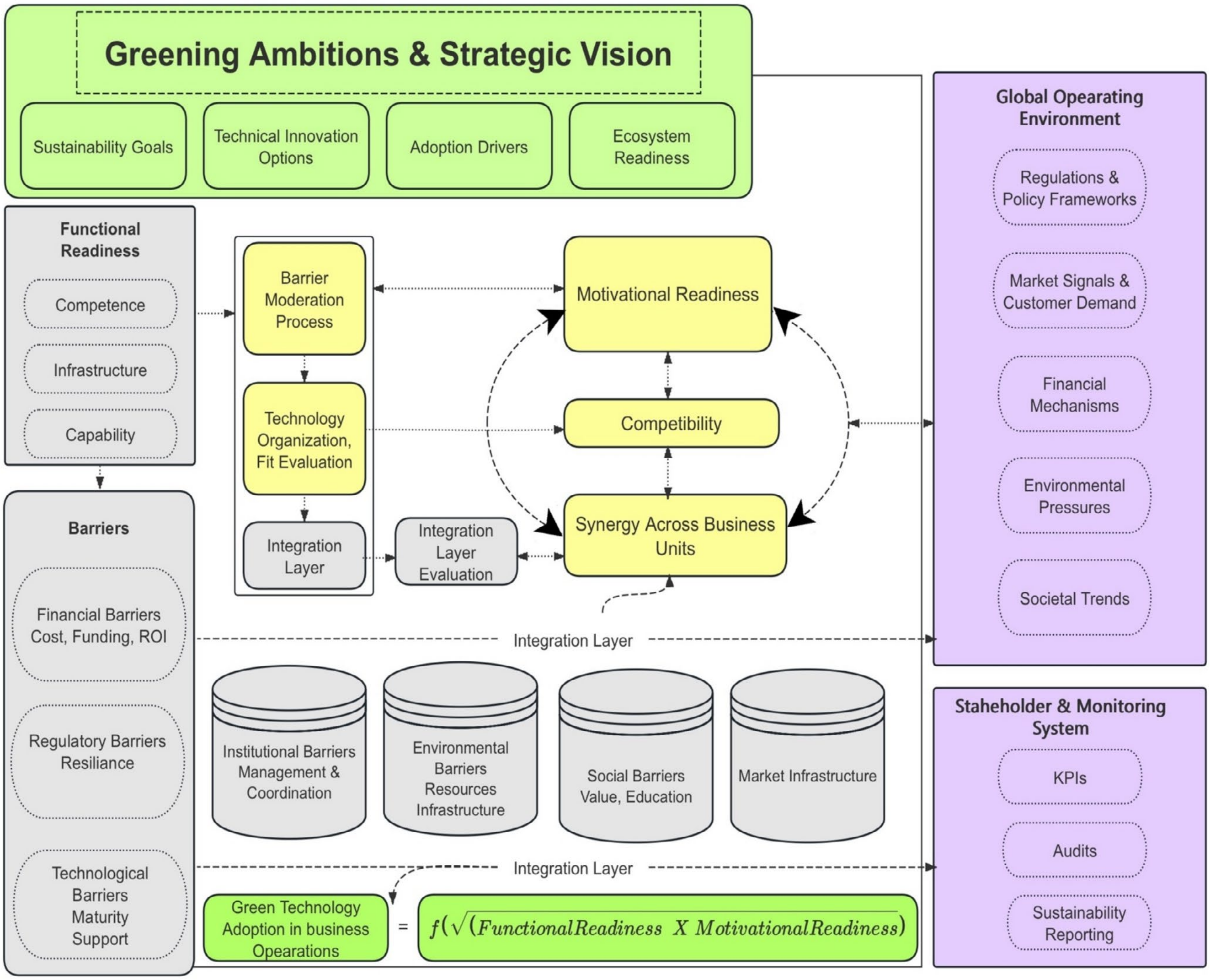

FIGURE 1 | An integreated green technology adoption barriers framework.

of literature-derived taxonomy and expert validation, lending credibility to the structure of our analysis. To our knowledge, no universally accepted guideline specifies how many experts are required for this type of analysis. Prior studies have employed different panel sizes to ensure robust results. For instance, Akusta et al. (2025) engaged seven experts to assess barriers to renewable energy for Turkey's sustainable energy planning, whereas Abdullah et al. (2022) used a five-member panel within an FAHP framework to identify strategic solutions to mitigate barriers for sustainable manufacturing. Guided by this precedent and balancing diversity with feasibility, we convened eight experts from academia and industry for the present study. Details of the expert panel are provided in the (Supporting Information: Table A1).

The experts were selected based on explicit criteria to ensure high relevance and credibility. Each expert had a minimum of 10 years of professional experience in fields related to sustainable technology and environmental policy. The panel was balanced between academia and industry: two members are university researchers specializing in green innovation and policy, and six are senior industry practitioners and government officials involved in green technology projects. All experts possess advanced knowledge of the specific barrier categories addressed in this study. These selection criteria and diverse expert profiles strengthen the validity of the input and align with recommended practices for expert-based MCDM studies.

3.1 | Fuzzy Delphi Method

Hsu et al. (2010) presented the fuzzy Delphi approach, which has since emerged as an established technique in decision-making research for the screening and structuring of evaluation criteria. The fuzzy variant of the Delphi procedure provides a more efficient method for consolidating expert opinions by accommodating the fundamental inconsistency of linguistic judgments and eliminating the need for multiple survey rounds. Conventional Delphi exercises often require several iterations before the responses converge, which can be time-consuming and may still result in weak consensus or response exhaustion among experts. The fuzzy Delphi method typically achieves a stable collective judgment in one iteration by representing experts' opinions as fuzzy numbers and mathematically aggregating them.

In fuzzy Delphi, the level of consensus among experts is expressed via a membership function. Prior studies have utilized various fuzzy number shapes, including triangular, trapezoidal, and Gaussian, to characterize this membership. Following this stream, the present study employs triangular fuzzy numbers (TFNs) to represent experts' assessments and to construct the membership degree function. The aggregated TFNs are subsequently combined through the geometric mean to identify the convergence point of the expert group. This study employs a three-phase procedure:

Phase 1: Aggregating individual fuzzy judgments.

Let the significance assigned to the ith component (where $k = 1$, 2, 3, ..., m) by the jth expert (where $j = 1, 2, 3, ..., n$) be represented as $w_{kj} = (akj, bkj, ckj)$. Subsequently, the fuzzy weight of the ith element ($k = 1,2,3, ..., m$) is denoted as $w_k = (ak, bk, ck)$, where the components ak, bk, and ck are calculated as follows

$$a_k = \mathrm{Min}\{a_{kj}\} = b_j = \frac{1}{n} = \sum_{k=1}^{n} b_{kj},\ c_k = \mathrm{Max}\{c_{kj}\} \tag{1}$$

Phase 2: Defuzzification of fuzzy weights.

During the second phase, the transformation of each fuzzy weight into an actual weight S_k by utilizing the center-of-gravity method:

$$S_k = \frac{\sum a_k, b_k, c_k}{3} \tag{2}$$

Phase 3: Selection of the definitive criteria set.

In the final step, the criteria for selecting GTA are established by comparing the weight of each barrier with the threshold $\tilde{a}$. The average of the weights assigned to all GTA criteria represents the threshold value a. If the weight value of GTA criteria is greater than or equal to $\tilde{a}$, then the criteria are selected. Otherwise, it will be rejected. In adopting this strategy, decision-makers' support for the FAHP technique was utilized to calculate the weights of the primary and sub-attributes of GTA using a pairwise comparison matrix.

3.2 | FAHP Method

The FAHP has been utilized to achieve optimal outcomes for complex decision-making processes, due to its enhanced accuracy relative to traditional models (Kumar et al. 2023). The FAHP is a term that refers to fuzzy set theory and advanced adjusted AHP, in which each membership component is represented by a membership function. Furthermore, each aspect of incorporation is analyzed in greater detail, categorized into levels that span from 0 to 1. Inability to participate is represented by a value of 0, whereas complete participation within the set is indicated by a value of 1. The aspect's degree of membership is quantified by any value that lies within these two extremities, indicating that it is partly incorporated into the set. The numbers that are fuzzy are essential as the basis for linguistic function membership to address uncertainties encountered in the valuation phase. TFNs are utilized to represent these linguistic values within the framework of the FAHP.

FAHP is recognized for its durable and robust attributes and is frequently employed in a variety of fields. The applications cover a variety of tasks, including the determination of the significance of social aspects in business sustainability (Arslan et al. 2023), the evaluation of the potential for groundwater in semiarid regions (Chen et al. 2025), the determination of green tourism sustainability (Li et al. 2024).

The FAHP framework consists of six stages for prioritising GTA barriers.

Stage 1: In this phase, the hierarchical framework of the issue is established.

Stage 2: Following the establishment of the hierarchical issue scheme, one of the most critical phases in the FAHP technique is to create a significance scale for relative importance comparisons. This rating mechanism, which spans from 1 to 9, employs TFNs to assist specialists in selecting investments in green technology based on numerical values. There exist five TFNs: $\tilde{1}$, $\tilde{3}$, $\tilde{5}$, $\tilde{7}$, and $\tilde{9}$. Table 2 and Figure 2 contain the listed TFNs. TFNs that efficiently gather and evaluate expert opinions mitigate uncertainty and confusion during the evaluation process. In order to facilitate a comprehensive assessment and understanding of the factors influencing decision-making.

Stage 3: To prioritize investments in GTA, the approach seeks to develop a fuzzy comparative matrix. Specialists employed TFNs to perform relative importance comparisons for this assessment. The arithmetic mean of the experts' contributions was employed to construct a fuzzy analysis matrix, referred to as matrix Z, which integrated these assessments. Equation (3) highlights the preceding approach by converting the collection of expert inputs into a comprehensive fuzzy evaluation tool, which allows for more logical decision-making in terms of green technology's significance hierarchy.

$$Z = \begin{bmatrix} 1 & \tilde{a}_{12} & \ldots & \ldots & \tilde{a}_{1n} \\ \tilde{a}_{21} & 1 & \ldots & \ldots & \tilde{a}_{2n} \\ \ldots & \ldots & \ldots & \ldots & \ldots \\ \ldots & \ldots & \ldots & \ldots & \ldots \\ \tilde{a}_{n1} & \tilde{a}_{n2} & \ldots & \ldots & 1 \end{bmatrix} \tag{3}$$

where $\tilde{a}_{ij} = 1$ if $i = j$ and where $\tilde{a}_{ij} = \left(\tilde{1}, \tilde{3}, \tilde{5}, \tilde{7}, \tilde{9}\right)$ or where $\left(\tilde{1}^{-1}, \tilde{3}^{-1}, \tilde{5}^{-1}, \tilde{7}^{-1}, \tilde{9}^{-1},\right)$. If i is not equal to j. The inverse value

TABLE 2 | Linguistic variables of fuzzy numbers.

Scale	Linguistic characteristic	TFNs value
1	Equal important	(1,1,3)
2	Moderate important	(1,3,5)
3	Strong important	(3,5,7)
4	Very strong important	(5,7,9)
5	Extreme important	(7,9,11)

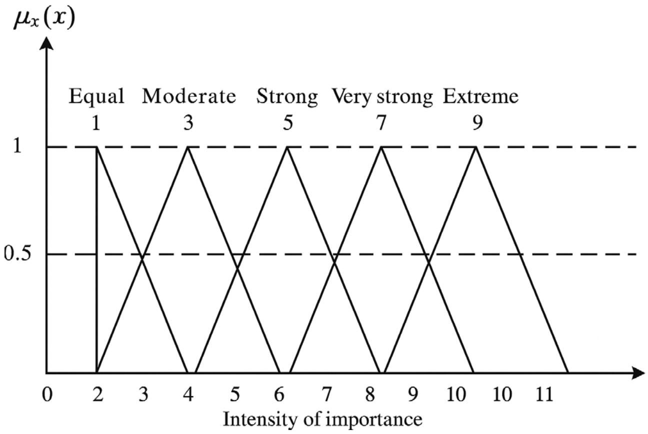

FIGURE 2 | The fuzzy membership function pertaining to linguistic variables.

for the scores is derived by inverting the Z matrix comparison to compute the score for each pair. Let $\widetilde{Z}_{ij}$ denote the value of matrix Z that represents the relations between components i and j.

$$\widetilde{Z}_{ij} = \frac{1}{\widetilde{z}_{ij}}.$$

Stage 4: This stage focusses on developing a detailed comparison matrix through the α-cut methodology to improve the accuracy of expert decision-making inputs. The α-cut method, introduced by Adamo in 1980, enables the management of fuzzy numbers by defining regions within the values that align with the α value. The α-cut is used to include only those fuzzy membership values above a certain threshold α, effectively focusing on the most plausible values and improving decision robustness. The optimism index (μ) represents the decision-maker's optimism level in aggregating fuzzy numbers. In this study, we set $\mu = 0.5$, reflecting a moderate level of optimism. This value was chosen to balance optimistic and pessimistic scenarios, consistent with common practice in fuzzy decision analysis, and was tested to ensure results were not overly sensitive to this parameter. Setting the value at $\alpha = 0.5$ and applying it to TFN results in an interval of a $0.5 = (2, 3, 4)$, as demonstrated in Figure 3.

The constant value in numbers of an is adjusted simply based on the projection level necessary to meet the requirements of the Z matrix, taking into account the optimism index μ obtained from the fuzzy comparison matrix. A comparison matrix for the α-cut can be constructed as follows:

$$\overset{\sim\alpha}{Z} = \begin{bmatrix} 1 & \overset{\sim\alpha}{b}_{12} & \dots & \dots & \overset{\sim\alpha}{b}_{1n} \\ b^{\alpha}_{21} & 1 & \dots & \dots & \overset{\sim\alpha}{b}_{2n} \\ \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots \\ \overset{\sim\alpha}{b}_{n1} & \overset{\sim\alpha}{b}_{n2} & \dots & \dots & 1 \end{bmatrix} \tag{4}$$

The μ denotes the degree of satisfaction or trust among stakeholders and functions as an efficiency metric within the selection matrix. This index measures decision-makers' faith in their decisions, with a higher score of μ indicating increased

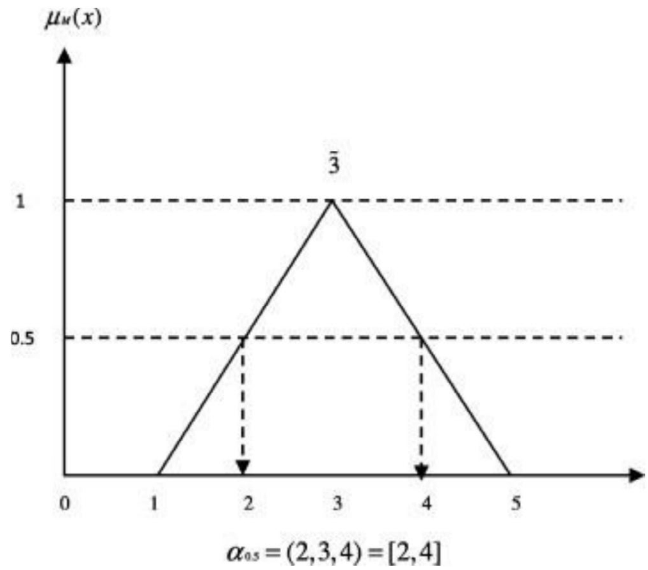

FIGURE 3 | TFN based α-cut operation (Wang, Li, and Liu 2023; Wang, Lyu, and Shen 2023).

satisfaction or confidence in the decision made. A linear concentric relationship is integrated through the procedure for calculating μ. This method quantifies the consensus or preference levels among decision-makers regarding specific alternatives within the decision-making framework, indicating their confidence and optimism in the assessment.

$$\overset{\sim\alpha}{b}_{ij} = \mu b^{\alpha}_{iju} + (1-\mu) b^{\alpha}_{iju} \text{ where } 0 < \mu \leqslant 1 \tag{5}$$

By adding μ to Equation (5), the fuzzy similarity matrix with α-cut becomes the sharp comparison matrix Z. The new matrix z is shown below:

$$Z = \begin{bmatrix} 1 & b_{12} & \dots & \dots & a_{1n} \\ b_{21} & 1 & \dots & \dots & \text{b} \\ \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots \\ b_{n1} & \text{b} & \dots & \dots & 1 \end{bmatrix} \tag{6}$$

Stage 5: Evaluating the consistency of results.

In order to validate the specialist's assessment, the coherence of each matrix and the overall homogeneity of the hierarchical structure were examined using a consistency ratio (CR). Consequently, the constant sharp comparison matrix Z is converted into a consistent fuzzy comparison matrix, Z'.

The following equation determines the matrix with the largest eigenvalue, which is given by Equation (7)

$$X_v = \lambda_{\max} v \tag{7}$$

where v indicates the main eigenvector value of the matrix.

The data obtained from the expert method's assessment is then verified by applying the CR for an approximate importance comparison. Equation (8) displays the CR assessment formula

$$\mathrm{CR} = \frac{\mathrm{CI}}{\mathrm{RI}} \tag{8}$$

$$\mathrm{CI} = \frac{\lambda_{\max} - k}{k - 1} \tag{9}$$

CI denotes the Consistem Index, while RI, as shown in Table 3, indicates the Random Index. The size of the underlying matrix is also denoted by the letter k.

If the relative importance comparison matrix's CR value is less than 0.10, the matrix is considered acceptable; if not, the experts' procedure must be repeated until the outcomes are more apparent.

Stage 6: Determine the weights assigned to the criteria.

Normalizing a portion or all of matrix Z's sections and rows in order to determine the weights related to green technologies is the last step in the FAHP development process.

3.3 | Fuzzy VIKOR Method

Opricovic first proposed the VIKOR technique in 1980 (Opricovic and Tzeng 2004). This strategy can assist decision-makers in reaching a final conclusion by calculating the weights of possibilities and ranking them to identify compromise solutions of a decision problem with incompatible or conflicting factors. A compromise is an agreement reached via reciprocal concessions, and in this case, the compromise solution is a workable option that is the closest to the ideal. The F-VIKOR levels are shown below:

Level 1: Performance-based fuzzy matrix development and the weight vector is as follows:

$$\widetilde{D} = \begin{array}{c} \\ O_1 \\ \vdots \\ O_n \\ \\ \end{array} \begin{bmatrix} C_1\, C_2 \cdots C_n \\ \widetilde{p}_{11}\widetilde{p}_{12} \cdots \widetilde{p}_{1n} \\ \widetilde{p}_{21}\widetilde{p}_{22} \cdots \widetilde{p}_{2n} \\ \vdots \; \vdots \; \vdots \; \vdots \\ \widetilde{p}_{m1}\widetilde{p}_{m2} \cdots \widetilde{p}_{mn} \end{bmatrix} \tag{10}$$

$$\widetilde{W} = (w_1, w_2, w_3), \; \sum_{j=1}^{n} w_j = 1$$

where O_i indicates the alternative i, that is $i = (1, 2, 3, \ldots, m)$; C_j represents the indicators j, that is $j = (1, 2, 3, \ldots, n)$; $\widetilde{p}_{ij}$ represents the performance of alternatives by fuzzy rating O_i with respect to indicators C_j; and $\widetilde{W}_j$ represent the fuzzy weight for each indicator. Thus, TFN can be written as $\widetilde{p}_{ij} = (x_{ij}, y_{ij}, z_{ij})$.

Level 2: Calculation of all benefit criteria values $\widetilde{p}_i^+ = (x_i^+, y_i^+, z_i^+)$ and the values of cost criteria $\widetilde{p}_i^- = (x_i^-, y_i^-, z_i^-)$. The set of benefit criteria is indicated as l^b whereas, l^c represent the set of cost criteria is followed as.

$$\begin{aligned} \widetilde{p}_i^+ &= \max_j \widetilde{p}_{ij}, \widetilde{p}_i^- = \min_j \widetilde{p}_{ij} \text{ for } i \in l^b \\ \widetilde{p}_i^+ &= \min_j \widetilde{p}_{ij}, \widetilde{p}_i^- = \max_j \widetilde{p}_{ij} \text{ for } i \in l^c. \end{aligned} \tag{11}$$

Level 3: Determine the values of normalized fuzzy decision matrix $\widetilde{D}_{ij}$:

$$\begin{aligned} \widetilde{D}_{ij} &= \frac{\widetilde{p}_i^+(-)\widetilde{p}_{ij}}{z_i^+ - l_i^-} \text{ for } i \in l^b \\ \widetilde{D}_{ij} &= \frac{\widetilde{p}_{ij}(-)\widetilde{p}_i^+}{z_i^- - l_i^+} \text{ for } i \in l^c. \end{aligned} \tag{12}$$

Level 4: Calculate the values $\widetilde{S}_j = \left(\widetilde{S}_j^x, \widetilde{S}_j^y, \widetilde{S}_j^z\right)$ and $\widetilde{R}_j = \left(\widetilde{R}_j^x, \widetilde{R}_j^y, \widetilde{R}_j^z\right)$:

$$\widetilde{S}_j = \sum_{i=1}^{n} \widetilde{W}_i(\times)\widetilde{D}_{ij} \tag{13}$$

$$\widetilde{R}_j = \max_i \widetilde{W}_i(\times)\widetilde{D}_{ij} \tag{14}$$

Level 5: Calculate the values $\widetilde{Q}_j = \left(\widetilde{Q}_j^x, \widetilde{Q}_j^y, \widetilde{Q}_j^z\right)$:

$$\widetilde{Q}_j = v\frac{\widetilde{S}_j(-)\widetilde{S}^+}{S^{-z} - S^{+x}}(+)(1 - v)\frac{\widetilde{R}_j(-)\widetilde{R}^+}{R^{-z} - R^{+x}} \tag{15}$$

Here $\widetilde{S}^+ = \min_j \widetilde{S}_j$; $S^{-z} = \max_j S_j^z$; $\widetilde{R}^+ = \min_j \widetilde{R}_j$; $R^{-z} = \max_j R_j^x$.

Additional, v is shown as a group strategy weight for most criteria $\widetilde{S}_j$; whereas, $(1 - v)$ is the individual weight of $\widetilde{R}_j$.

Level 6 and 7: Perform defuzzification on the values $\widetilde{S}_j$,$\widetilde{R}_j$ and $\widetilde{Q}_j$ values, respectively. The alternatives are prioritized in descending order based on the crisp values of S, R, and Q. The outcomes are three prioritization results $\{O\}_S$,$\{O\}_R$ and $\{O\}_Q$, respectively.

Level 8: To meet two subsequent conditions, propose a solution to the alternative $O^{(1)}$ which represents the optimal solution of the measure Q:

Condition 1: Required benefit Ben must be greater than or equal $\geq DQ$. where ben $= \frac{[Q(O^{(2)}) - Q(O^{(1)})]}{[Q(O^{(m)}) - Q(O^{(1)})]}$ is the benefit rate of alternative $O^{(1)}$ is prioritized over the alternative with the second-ranked $O^{(2)}$ in $\{O\}_Q$ and the threshold $\mathrm{DQ} = \frac{1}{(m-1)}$.

Condition 2: Stability of decision-making: Option $O^{(1)}$ ought to be regarded as the best option according to S or Q values. This should meet at least one of the CI and C2 requirements. A set of comprising solutions is proposed in the event that both conditions are not met, and they can be described as follows:

Condition satisfied C_1: Alternative $O^{(1)}$ and $O^{(2)}$ are applicable only if C_1 is not satisfied, or: only (C_2) is not satisfied, or:

Condition satisfied C_2: Alternative $O^{(1)}$, $O^{(2)}$, ..., $O^{(M)}$ if (C_1) is not satisfied; $O^{(m)}$ is defined by the following relation:

TABLE 3 | Random index.

k	1	2	3	4	5	6	7	8
RI	0	0	0.52	0.89	1.11	1.25	1.35	1.40

$\frac{[Q(O^{(M)}) - Q(O^{(1)})]}{[Q(O^{(m)}) - Q(O^{(1)})]} < DQ$ determine the optimal value of M. The alternatives are located in proximity to one another. The schematic methodological framework of GTA barriers ins presented in Figure 4.

4 | Results and Discussion

The findings of the paper are organized into three sections. The first section represents the inclusion/exclusion criteria of GTA barriers based on the Fuzzy Delphi method. The second section determines the most significant GTA barriers and sub-barriers; the FAHP is initially used. The subsequent approach, the Fuzzy VIKOR, provides a structured strategy to conquer these barriers by evaluating and ranking strategic alternatives presented in the third section. The findings offer an in-depth understanding of the main barriers and the best approaches to support a long-term shift to green technologies. The details of GTA barriers and sub-barriers are presented in Figure 5.

4.1 | Fuzzy Delphi Results

The fuzzy Delphi analysis determined a cut-off value of 5.125 based on the mean score of the GTA barriers. Barriers with scores at or above this threshold were retained, whereas those scoring below were excluded from further consideration. Table 4 summarizes that 11 GTA barriers met the requirement, whereas a limited set of GTA barriers, namely administrative burden barriers, expertise barriers, and insufficient cost benefit/barriers, failed to meet the threshold and were consequently excluded. Their exclusion does not indicate that these factors are unimportant in broader discussions on adoption of green technology; instead, it signifies that they fall outside the specific evaluative focus of this study. Experts consulted in this research opted to exclude it, contending that inadequate coordination among institutions would exert a uniform, system-wide influence on all GTA in developing countries, rendering it conceptually inappropriate to estimate its proportional impact on individual technologies.

4.2 | Fuzzy AHP Methods Results

In the first phase of this study, the FAHP method was applied using a relative importance matrix as the starting point. This matrix was built through the geometric average algorithm, which made it possible to calculate each expert's rating separately. Through this process, seven main barriers were identified, and each of them included three to five sub-barriers. Together, these levels formed a clear structure for examining the key obstacles to GTA. The sub-barriers helped to refine the evaluation by breaking each main category into more specific issues that allowed for a deeper and more systematic analysis. This hierarchical approach improved the accuracy of the results and provided a

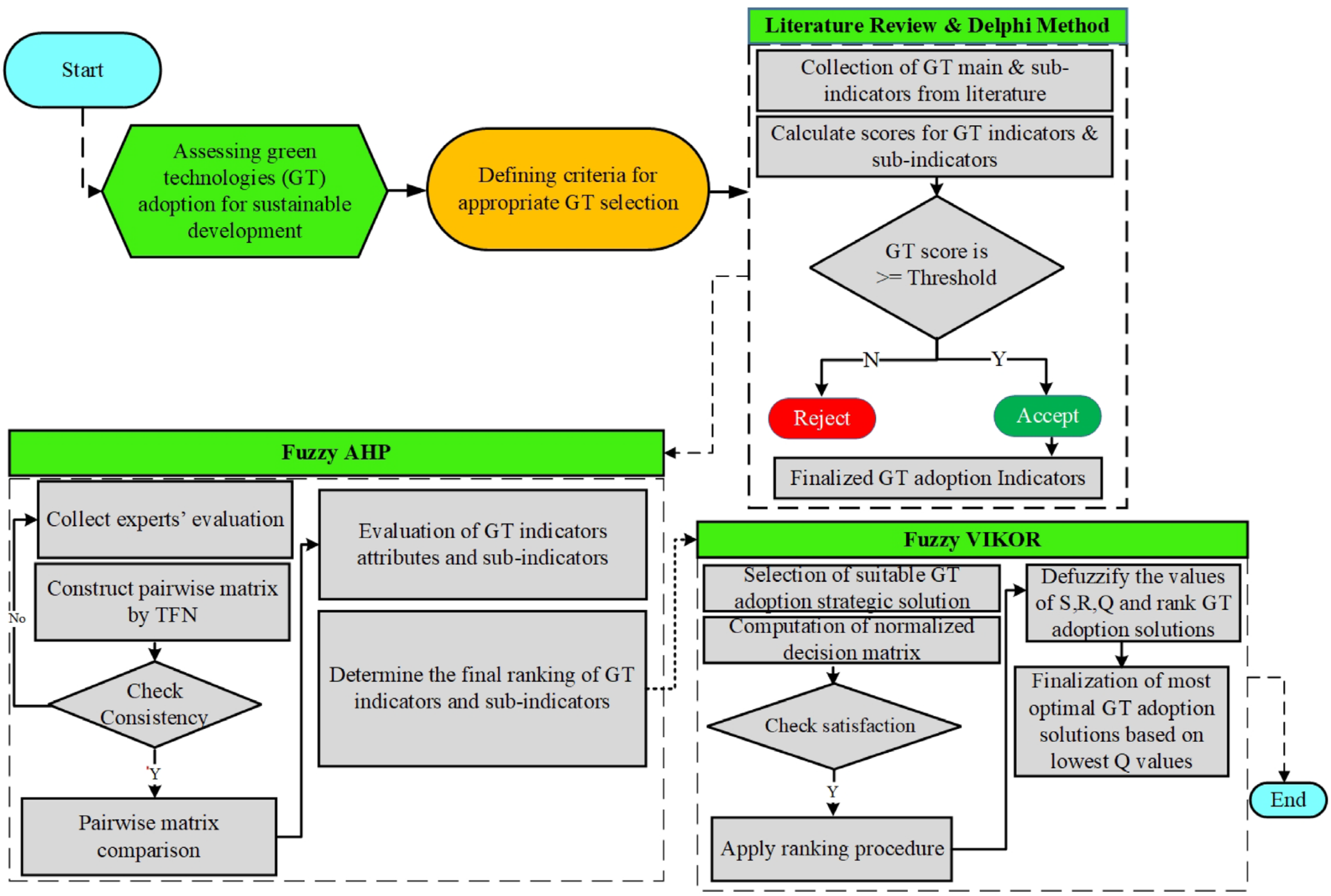

FIGURE 4 | Integrated fuzzy AHP and Fuzzy VIKOR decision support framework.

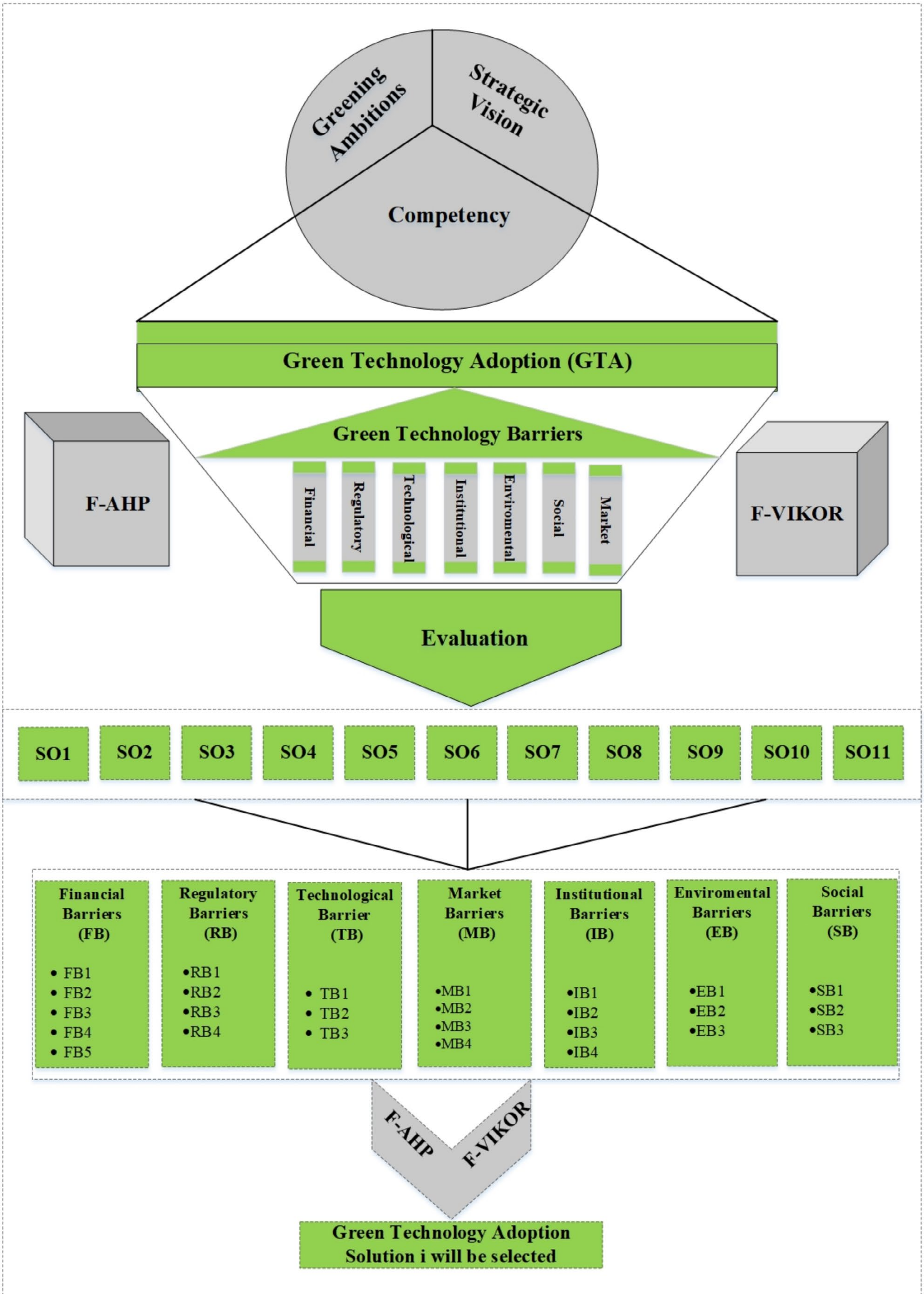

FIGURE 5 | The strategic framework of GTA barriers and sub-barriers.

TABLE 4 | Fuzzy Delphi results to finalize GTA barriers.

GTA barriers	*a*	*b*	*c*	Fuzzy Delphi scores	Selected √/rejected×	Barrier code
Financial barriers	3	6.8	9.9	6.57	√	FB
Regulatory barriers	3	6.7	9.8	6.54	√	RB
Technological barriers	3	6.2	9.3	5.83	√	TB
Market barriers	3	6.0	9.5	6.17	√	MB
Institutional barriers	3	7.6	9.8	6.80	√	IB
Environmental barriers	3	6.5	9.4	6.77	√	EB
Social barriers	3	5.9	9.7	6.21	√	SB
Administrative burden barriers	2	3.8	5.5	3.77	×	AB
Expertise barriers	2	4.1	5.4	3.83	×	EB
Insufficient cost benefit/barriers	2	4.3	5.3	3.87	×	CB

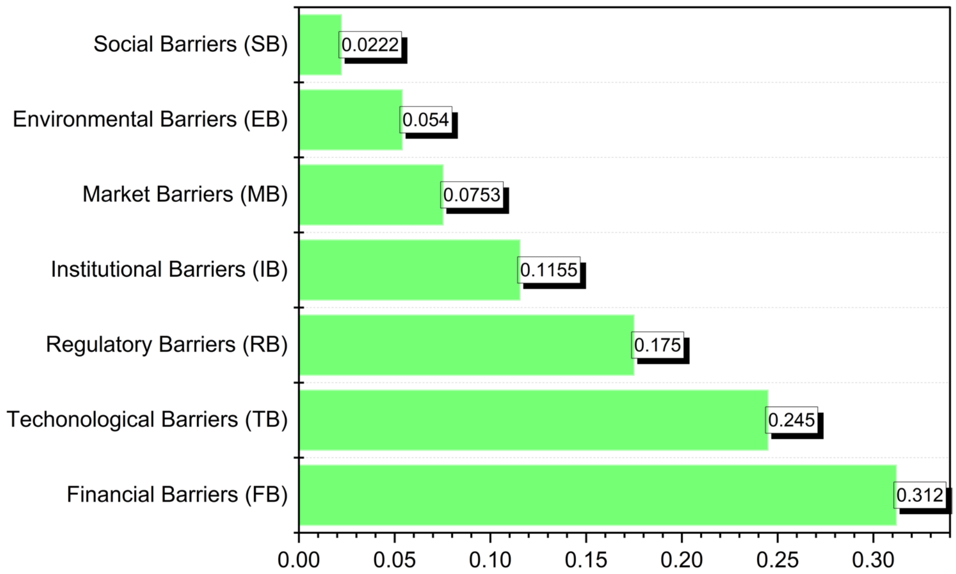

FIGURE 6 | Ranking of main GTA barriers.

broader understanding of the factors that limit sustainable technology development. The FAHP analysis was carried out in two main steps. First, the weights of the seven primary criteria related to GTA barriers were determined and integrated into the AHP framework. In the second step, the weights for the sub-criteria were calculated to offer a more detailed view. The final ranking of the main criteria for GTA barriers is presented in Figure 6. The fuzzy pairwise matrix of main green technologies adoption barriers is presented in (see Supporting Information: Table A3).

4.2.1 | The Ranking of Main GTA Barriers

Figure 6 presents the weights of each primary criterion derived through the FAHP method. Financial Barriers (FB) emerge as the most critical challenge among the seven identified barriers, carrying the highest weight of 31.20%, underscoring the fundamental role of financial constraints in hindering the transition to sustainable technologies. Technological Barriers (TB) rank second, with a weight of 24.50% followed by regulatory barriers (RB) in the third position at 17.50%. Institutional barriers (IB) rank fourth at 11.55%, followed by market barriers (MB) at 7.53%, environmental barriers (EB) at 5.40%, and Social Barriers (SB) at 2.22%.

The implementation of green technology in developing economies is being slowed down by financial hurdles, which make up 31.20% of the total weight. One of the primary obstacles is the restricted availability of credit and investment capital, which makes it challenging for entrepreneurs and enterprises to fund sustainable projects. Approximately 84% of developing economies' climate and green technology entrepreneurs rely on self-financing in the early phases of their businesses (Soumonni and Ojah 2022). This circumstance demonstrates the acute lack of outside finance and the pressing need for more robust financial systems to assist the nation's green transition (The world bank 2023). The limited availability of financing continues to discourage many potential investors from engaging in green innovation. Technological challenges appear as the second major obstacle to adopting green technologies largely because access to advanced tools and systems remains restricted in several sectors.

In agriculture and other key industries, many actors still depend on conventional methods, and the use of modern, climate-smart technologies is progressing only slowly. Despite ongoing efforts to enhance irrigation efficiency through drip irrigation systems, the widespread implementation of such innovations remains constrained. This limitation impedes optimal water resource management and reduces agricultural productivity, highlighting the urgent need for broader technological dissemination and support (The world bank 2023).

4.2.2 | Financial Sub Barriers

Figure 7 presents the ranking of financial barriers' sub-criteria as follows: FB1 > FB4 > FB3 > FB2 > FB5. The FB1 refers to the lack of access to financial resources with a weight of (32.15%) followed by the Unavailability of Government Subsidies with a weight of (25.83%). This finding echoes a commonly recognized issue in the literature, where restricted access to funding is consistently emphasized as a primary barrier to deploying green technologies, especially among developing nations where there is still restricted access to long-term, affordable funding. According to IRENA (2023), concessional financing comprised only 1% of all renewable energy financing in 2020, restricting businesses' capacity to undertake the high-capital projects required for a sustainable energy transition, particularly in developing nations. Our analysis found a continuing gap in financial accessibility despite the existence of financing tools, supporting the notion that FB1 and FB4 are the main barriers to the adoption of green technologies. UNDP (2023) pointed out that public funds are frequently inadequate and poorly allocated, particularly for distributed or small-scale projects, even though it is crucial for early-stage green investments. Green technology implementation is slowed down by the absence of steady, predictable government assistance, which discourages private sector involvement and increases overall investment risk. Developing economies may be able to overcome this obstacle by strengthening current green finance structures, rendering them more accessible to all sectors through better regulatory clarity, and boosting risk-mitigation strategies to attract private investment.

Unattractive returns on investment (FB3) and high-cost premium (FB4) are ranked third and fourth with a respective score of 20.12% and 14.97%. Subsequently, high operational and maintenance costs (FB5) are the least impactful sub-criteria among all with a weight of 6.9%.

4.2.3 | Technological Sub-Barriers

The sub-criterias of technological barriers are presented in Figure 8 and ranked as follows: TB2 > TB3 > TB1, the most impactful sub-barrier is Obsolete Infrastructure with a weight of (41.25%). These findings align with global patterns of energy investment in developing economies. Despite increased global urgency and the relative economic viability of green energy sources, clean energy spending in developing nations, especially in Africa and parts of Asia, is still at 2015 levels (Wang, Pal, et al. 2025).

Research on the development of green banking products in Pakistan has identified challenges including inadequate technological expertise, limited awareness of green financial innovations, difficulties in pinpointing target markets, and insufficient access to funding mechanisms designed for green initiatives (Khan et al. 2024). A qualitative study on green banking in India identified numerous obstacles associated with the adoption of green technology, unwillingness to embrace new digital and environmental technologies, insufficient awareness of green tools and services, along with elevated short-term implementation costs and technical constraints (Sharma and Choubey 2022). These insights highlight the crucial need for technology development, strategic investment in green technologies, and the development of human resources to facilitate a quicker green transition.

4.2.4 | Regulatory Sub-Barriers

The four criterions of regulatory barriers are presented in Figure 9 and ranked as follows: RB4 > RB1 > RB3 > RB2. The two most impactful sub barriers are Conmplex and rigid legislation (RB4) with a weight of (34.56%) and Unclear regulations (RB1) with a weight of (29.52%). MENA economies, such

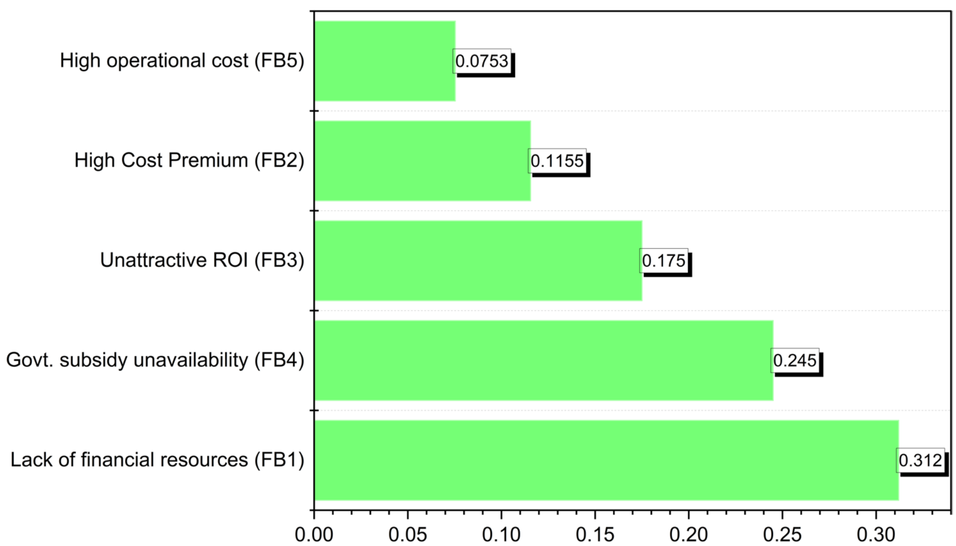

FIGURE 7 | Ranking of the financial sub-barriers.

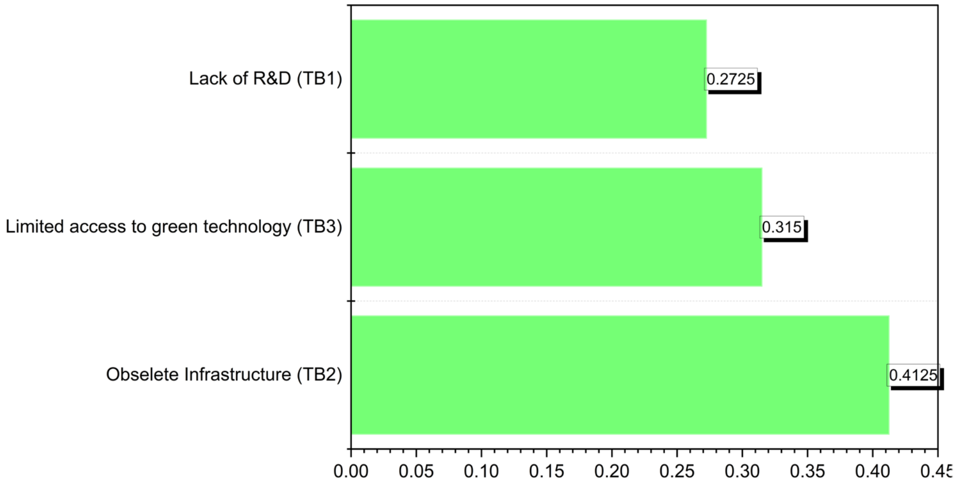

FIGURE 8 | Ranking of the technological sub-barriers.

as Morocco, experience delicate administrative processes and obsolete legal structures that lack adaptability. These inflexible frameworks obstruct private sector involvement in green energy, notwithstanding the region's abundant solar and wind resources. It additionally emphasizes that reforms in energy price and regulation are essential for facilitating investment and sustainably expanding renewable energy (Liu et al. 2023). Ambiguous and contradictory governmental objectives and regulatory structures are primary impediments to the progress of green technology implementation. Moreover, insufficient interagency coordination, ambiguous licensing procedures, and the lack of persistent legal assurances impede investor trust and operational planning (Owojori and Erasmus 2025).

Ineffective enforcement of environmental laws (RB3) and lack of government support/incentives (RB2) are ranked third and fourth with a weight of (21.51%) and (14.41%) respectively. Regulatory measures are crucial in the green energy ecosystem, for they protect financial stability by reducing market imbalances and promoting enhanced openness, uniformity, and predictability in investment implementation. Nauman et al. (2024) assert that stable and consistent rules are essential for attracting long-term capital, particularly in emerging nations where green technologies contend with substantially.

4.2.5 | Institutional Sub-Barriers

Poor coordination (IB2) has ranked as the most critical sub-barrier within the Institutional Barriers affecting the development of green technologies with the highest weight value of (34.25%) (see Figure 10). This issue is particularly evident in the overlapping responsibilities between national bodies and regional authorities, which often leads to project delays and inefficient implementation of green strategies. To streamline planning, funding, and deployment processes for green technologies effective coordination among these entities is primordial. The remaining sub-barriers are prioritized as follows: Bureaucratic inefficiency (IB1) ranks as the second most significant barrier with a weight of (29.58%), indicating that complicated administrative procedures and long-lasting approval processes persist in preventing both domestic and foreign investment (Zhang and Li 2022).

Limited institutional capacity (IB3) ranked in the third position within the category obtaining a weight of (26.04%), indicating a deficiency in technical understanding and insufficient human resources within essential energy-related organizations. Ultimately, political instability (IB4) is positioned fourth, possessing the minimal weight of (10.13%). Investor attitude is significantly influenced by domestic and global political and economic developments. The geopolitical tensions and economic instability in the MENA area highly affect the returns and volatility of green technologies (Ma and Najam 2024). This sensitivity underscores the necessity of preserving a stable legislative framework and executing proactive risk management techniques to protect investor trust and guarantee the continued expansion of green technologies investment in the country.

4.2.6 | Market Sub-Barriers

Market barriers sub-criterias are ranked as follows: MB2 > MB1 > MB4 > MB3. As represented in Figure 11. Limited Awareness of Green Technologies (MB2) recorded the most significant score of (33%) within its category. These results are in line with (Zhao, Teng, and Ji 2024; Zhao, Gao, et al. 2024), who stress the importance of public education on the monetary and environmental advantages of green technologies as a means to overcome the current information gap and break into the market. The following sub barrier is the low demand for sustainable products (MB1) received the weight value (29.50%). Whereas, unequal distribution of benefits (MB4) comes in row on the third position within the market barriers category with a weight of (22%); this sub barrier highlights systemic equity issues; wherein small firms may regard themselves as marginalized from the actual or perceived advantages of the green transition. Lastly, market resistance to change (MB3) is the lowest ranked of the four with a weight of (15.50%) indicating that, although cultural stagnation and a preference for traditional practices are still factors, they are

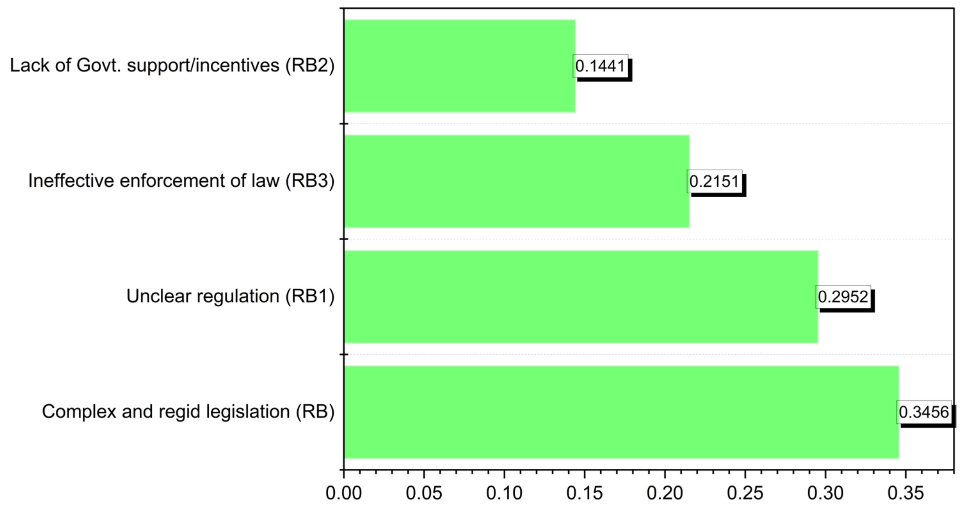

FIGURE 9 | Ranking of the regulatory sub-barriers.

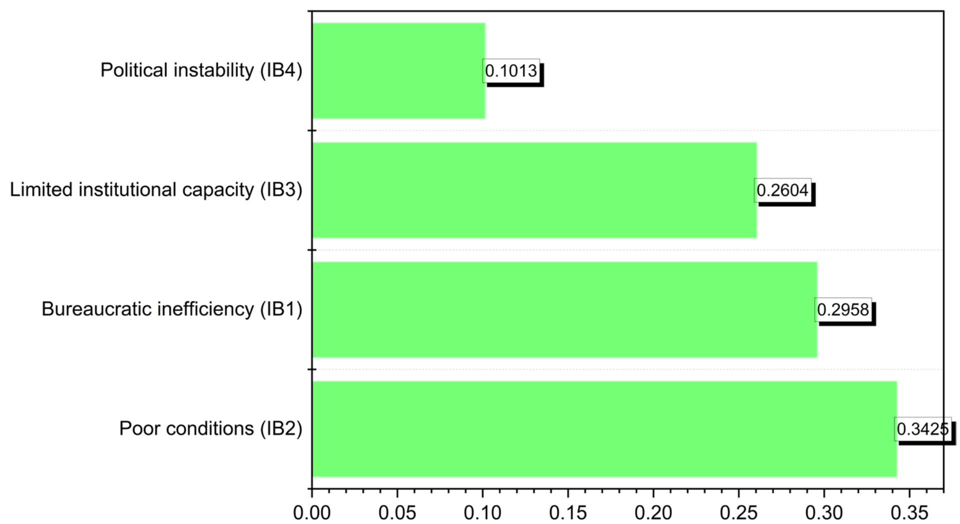

FIGURE 10 | Ranking of the institutional sub-barriers.

not the primary roadblocks that hinder progress. Collectively, these market-related results emphasize the crucial importance of consumer-oriented incentives, inclusive policy formulation, and public awareness campaigns to expedite the widespread use of green technologies in developing countries.

4.2.7 | Environmental Sub-Barriers

Among the environmental barriers (EB), climate challenges (EB2) have been recognized as the most significant sub-barrier to the advancement of green technologies, possessing the greatest weight value of (38.19%). Developing countries such as Morocco and Egypt are most environmentally vulnerable in the region due to harsh weather phenomena, including droughts, desertification, and erratic precipitation patterns, considerably affecting the viability and dependability of green technology initiatives (Ikram 2025). Subsequently, natural resource dependency (EB1) is ranked second with a weight value of (34.00%). Finally, limited natural resource availability (EB3) is ranked third, with a weight value of (27.81%) as presented in Figure 12.

4.2.8 | Social Sub-Barriers

Social Inequalities (SB3) have ranked as the most important sub-barrier within the social barriers (SB) category, obtained the highest weight value of (40.62%), is presented in Figure 13. Whereas, low stakeholder engagement (SB2) is identified as the second most significant obstacle, assigned a weight value of (35.10%). Numerous renewable energy initiatives in developing economies have encountered obstacles due to insufficient engagement of local groups, commercial sector participants, and civil society organizations during the design and execution phases (Brito Cedeno and Wei 2024). Moreover, public resistance to change (SB1) is ultimately rated third, assigned a weight value of (24.78%). Switching to renewable energy frequently necessitates major shifts in consumption behaviors, land execution, and societal norms.

4.2.9 | Overall GTA Sub Barriers Results

The final ranking of sub-criteria, as illustrated in Figure 14, is established by multiplying the global weights of the

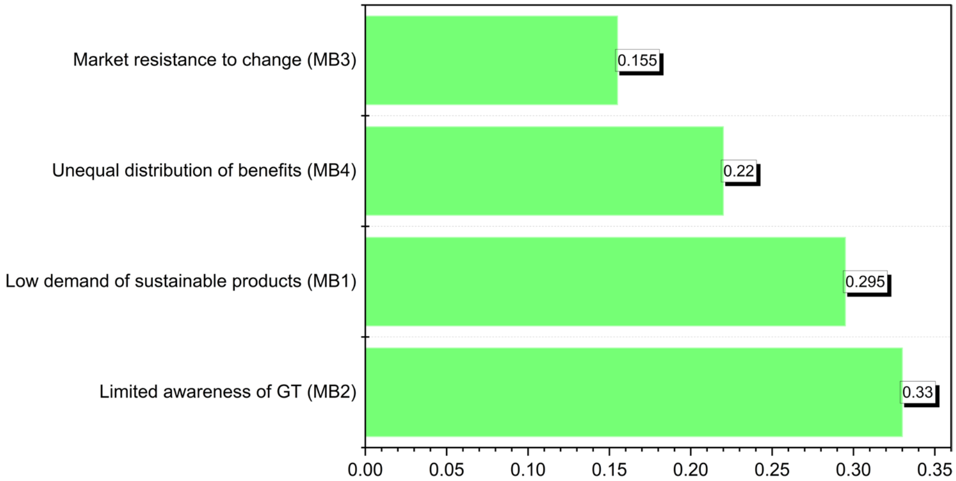

FIGURE 11 | Ranking of the market sub-barriers.

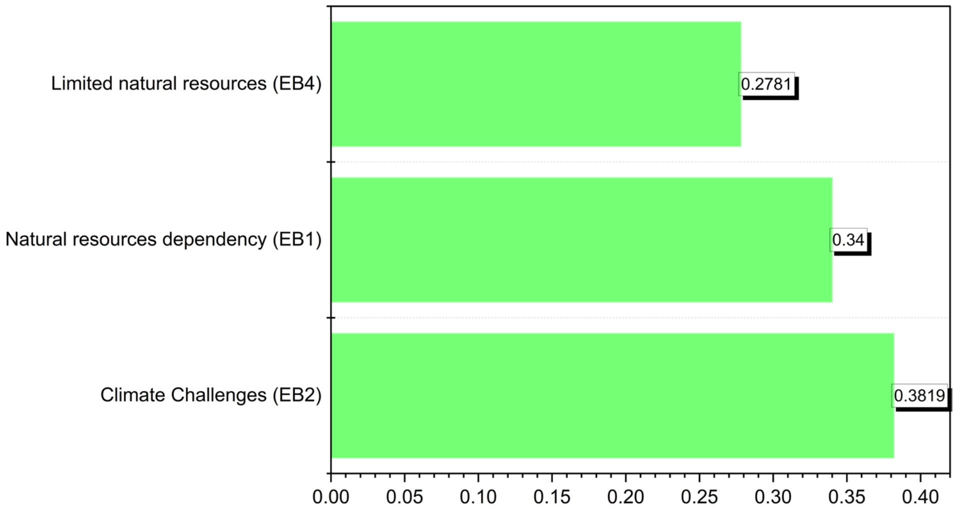

FIGURE 12 | Ranking of the environmental sub-barriers.

sub-criteria by the weight of each respective category. The overall ranking of GTA barriers classification is determined by the final weights assigned to all sub-criteria. The attributes of sub-criteria for classifying green technologies in developing economies are presented as follows: EB1, IB3, EB2, SB1, IB2, EB3, IB4, IB1, FB1, RB2, SB2, FB4, RB3, MB4, FB5, MB1, TB1, RB1, MB2, SB3, FB2, RB4, TB2, MB3, TB3, FB3. In this study's classification of GTA barriers, Natural resource dependency (EB1) is the sub-criteria within the environmental category that received the highest weight score of 0.0734, followed by Limited institutional capacity (IB3) with a weight of 0.0722. Climate challenges (EB2) and public resistance to change (SB1) occupy the third and fourth positions, respectively, with weight scores of 0.0691 and 0.0668, as shown in Table 5. The final three sub-barriers assessed for the classification of GTA in developing economies are market resistance to change (MB3) 0.0126, limited access to green technologies (TB3) 0.0104, and unattractive returns on investment (FB3) 0.0075, respectively. The fuzzy pairwise matrix of GTA sub-barriers is presented in (see Supporting Information: Tables A4–A10).

4.3 | Fuzzy VIKOR Results

To ascertain which solutions are optimally situated to mitigate the barriers to GTA, the FVIKOR method helped assess alternatives through three principal dimensions: overall group utility S_i, individual regret R_i, and a compromise ranking index Q_i. Among these, Q_i is the most pivotal, as it merges both the collective advantage and the worst-case performance and assists in prioritizing the most favorable policy alternatives, as shown in Table 6. Moreover, the Fuzzy VIKOR matrix is presented in (see Supporting Information: Table A11).

Each solution was assigned a weight Q value where lower weight suggests enhanced overall performance and greater balance across different evaluation criteria. The most favorable solution is Resilience programs for green tech jobs, deemed (SO7) with the lowest score of 21.46% (see Figure 15). This result proves that the most balanced and effective strategy for facilitating the adoption of green technology is to invest in workforce training and growth. As such, investments provide professionals with the necessary skills for the

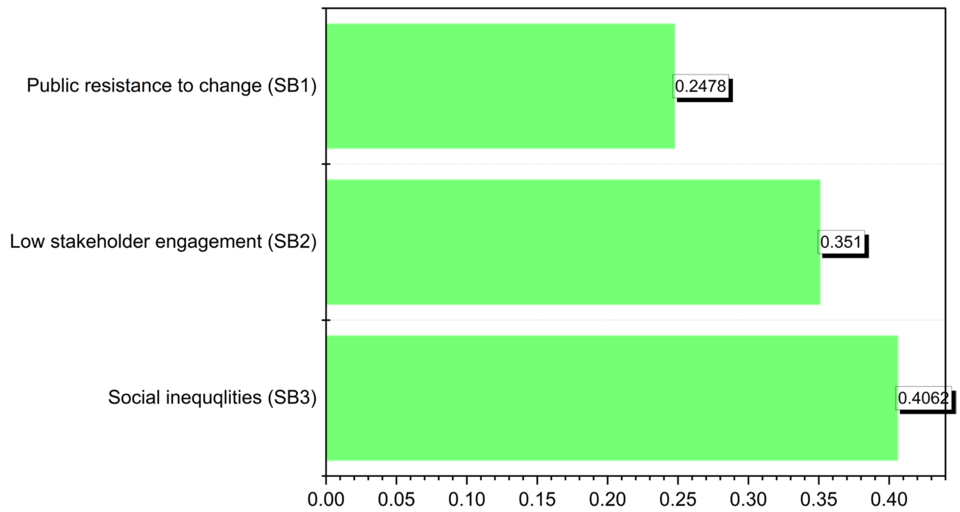

FIGURE 13 | Ranking of the social sub-barriers.

emergence of green sectors and promote a seamless transition to a green economy. Its first ranking makes it a highly practical and inclusive solution for overcoming green technology barriers. The second crucial solution is banks consolidating digital tools for green investments, with a weight of 24.46%. SO6 endorses the use of contemporary financial instruments to optimize investment flows toward green projects, enhancing the accessibility and efficiency of funding, especially for SMEs and startups. The third crucial solution is a platform connecting vendors and financial institutions (SO4), with a *Q* value of 29.72% (see Table 7). SO4 stresses enhancing collaboration between green technology innovators and the financial institutions. In most developing countries, several green technology ventures encounter barriers not necessarily due to insufficient finance but due to inadequate connections between the supply and financing system. A digital or an institutional specialized platform can facilitate the connection between technology innovators, service providers, banks, investors, and funding programs. Its high ranking shows its potential to unlock funding and accelerate the adoption of green technologies across the country. Solutions Simplifying and modernizing regulatory processes (SO2) and Coordinated efforts between key institutions (SO10) appear to also have a primordial impact, ranking fourth and fifth, with a score of 35.32% and 39.61%, respectively (see Figure 16). These are crucial for guaranteeing that Morocco's institutional environment facilitates green development. Minimizing bureaucratic obstacles and enhancing collaboration among ministries and governmental agencies may expedite project approvals and prevent policy fragmentation.

Strategies with significant Q_i values, such as Capital subsidies to reduce initial costs (SO11) and Green industrial policies for targeted support (SO1), although crucial, are deemed less immediately efficient according to the current judgment.

The FVIKOR approach prioritizes strategies for converting intricate green technology barriers into implementable paths, guiding stakeholders to focus on the most impactful and feasible solutions. His systematic method guarantees that resources and efforts are allocated to solutions with the highest potential for enduring impact, meeting both green growth objectives and economic requirements.

This study's findings, along with a thorough evaluation of recent advances in developing economies' green technology sector, offer essential insights into the primary and secondary obstacles to GTA. The Fuzzy AHP results unequivocally indicate that financial barriers (FB), technological barriers (TB), and regulatory barriers (RB) are the most substantial impediments, corroborating prior literature. This conversation primarily highlights the importance of our study in academia, examining its practical consequences for policymakers, industry stakeholders, and the general public. The highest-ranked solutions; reskilling programs for green tech jobs (SO7) and Banks consolidating digital tools for green investments (S06) indicate that solutions targeting institutional and financial barriers are perceived as most urgent. Moreover, the highest-ranked strategy, resilience programs for green tech jobs (SO7), tackles the critical skills deficit. Experts prioritized this because a well-trained workforce is fundamental to implementing new technologies and sustaining green initiatives, thereby amplifying the impact of subsequent investments (OECD 2023). Moreover, the integration of sustainability into educational and urban systems through structured knowledge management not only enhances environmental awareness but also accelerates behavioral shifts critical to the long-term adoption of green technologies (Basilico, Biancardi, et al. 2025; Basilico, D'Adamo, et al. 2025). In contrast, while direct capital subsidies (SO11) are valuable for lowering financial hurdles, they were viewed as slightly less foundational; funding without human capital and institutional capacity may not yield lasting progress. This synergistic rationale explains why capacity-building strategies outrank purely financial interventions in our results. Each top-ranked solution thus addresses a leverage point in the system of barriers, which, if strengthened, produces broad positive effects on the adoption of green technology.

This finding aligns with broader international experiences where human capital development and financial innovation are recognized as crucial enablers for green transitions (Abbas and Najam 2025). The identification of financial barriers as a

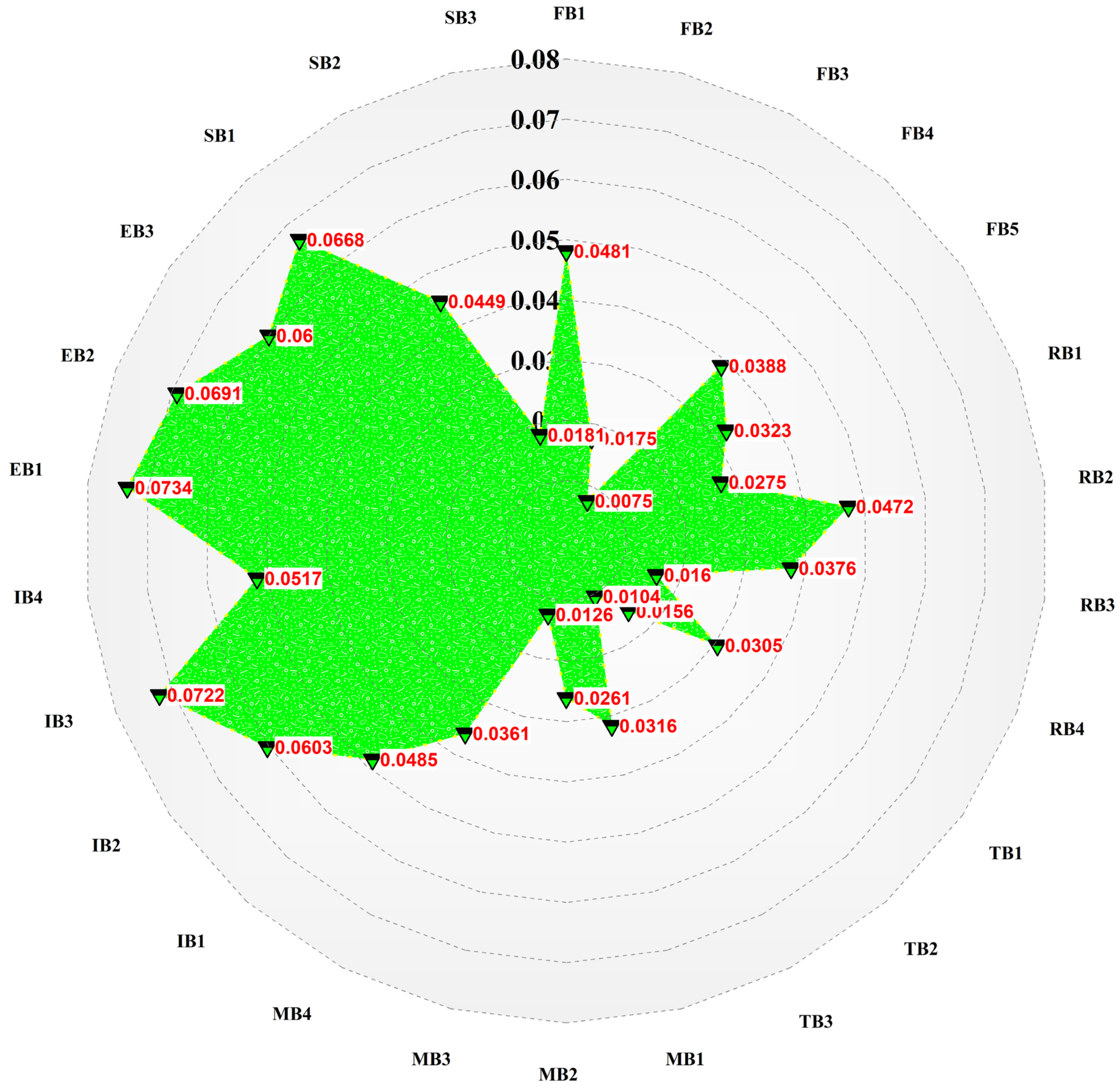

FIGURE 14 | Overall ranking of GTA sub-barriers.

primary limitation aligns with the issues of Morocco's green revolution highlighted in recent reports in which the accessibility of affordable finance for renewable energy and green infrastructure persists as a constraint despite the nation's ambitious and optimistic renewable energy objectives. The economic structures such as energy communities can serve as pivotal enablers of green technology diffusion by addressing local investment barriers and aligning stakeholder incentives, especially in resource-constrained settings (Basilico, Biancardi, et al. 2025; Basilico, D'Adamo, et al. 2025). Sub-barriers such as lack of access to financial resources (FB1) and unavailability of government subsidies (FB4) scored high in their respective category indicating systemic limitations in financial markets and an urgent need for more substantial state-supported incentives.

The focus on streamlining and modernizing regulatory processes (SO2) underscores the regulatory challenges identified in both the present study and previous scholarly works (Tian and Pang 2023). In this setting, legislative reforms are essential rather than merely supplementary, serving as the foundation for all subsequent policy activities. Moreover, the barrier hierarchy observed here is broadly reflective of challenges in many developing economies. The recent studies have identified financial constraints and regulatory gaps as top barriers to green initiatives in countries such as India and Malaysia (Wasan et al. 2021; Majekodunmi et al. 2023). This suggests that Morocco's experience is not isolated; limited access to capital, insufficient technology infrastructure, and policy enforcement issues are common impediments across emerging economies. Therefore, the strategic solutions prioritized in our study include dedicated green financing mechanisms, strengthening regulatory frameworks, and capacity-building programs which are applicable to other developing nations facing similar sustainable development constraints. By generalizing our findings, this study provides a roadmap that policymakers in various developing country contexts can adapt to accelerate GTA.

TABLE 5 | Final ranking of GTA barriers.

Main barriers	Main barrier weights	Sub-barriers codes	Consistency ratio (CR)	Global weight	Local weight
Financial barriers	0.3120	FB1	0.0335	0.0481	0.0250
		FB2		0.0175	0.0241
		FB3		0.0075	0.0266
		FB4		0.0388	0.0415
		FB5		0.0323	0.0555
Regulatory barriers	0.1750	RB1	0.0370	0.0275	0.0389
		RB2		0.0472	0.0197
		RB3		0.0376	0.0162
		RB4		0.0160	0.0231
Technological barriers	0.2450	TB1	0.0431	0.0305	0.0549
		TB2		0.0156	0.0528
		TB3		0.0104	0.0511
Market barriers	0.0753	MB1	0.0444	0.0316	0.0324
		MB2		0.0261	0.0637
		MB3		0.0126	0.0299
		MB4		0.0361	0.0113
Institutional barriers	0.1155	IB1	0.0179	0.0485	0.0393
		IB2		0.0603	0.0631
		IB3		0.0722	0.0363
		IB4		0.0517	0.0194
Environmental barriers	0.0540	EB1	0.0172	0.0734	0.0573
		EB2		0.0691	0.0330
		EB3		0.0600	0.0651
Social barriers	0.0222	SB1	0.0862	0.0668	0.0125
		SB2		0.0449	0.0649
		SB3		0.0181	0.0425

5 | Conclusion and Policy Implications

This study intended to investigate the barriers impeding the adoption of green technologies in Morocco and to offer pragmatic solutions that would expedite the nation's transition to sustainability. Using a hybrid decision-making framework that included the Fuzzy VIKOR approach and the Fuzzy Analytic Hierarchy Process (FAHP), the study offered a methodical evaluation of the most significant challenges and the most effective strategies to overcome them. The findings of this research offer valuable insights for policymakers, financial institutions, industry stakeholders, and researchers concerned with advancing green transitions in emerging markets.

According to the FAHP assessment, the most substantial hurdle to green technology adoption in Morocco is financial restrictions, which account for 31.2% of the aggregate. The most pressing sub-barriers in this category were a lack of access to financial resources and the absence of government subsidies, indicating structural flaws in Morocco's financial ecosystem. Technological barriers, which ranked second at 24.5%, were mostly related to outdated infrastructure and limited access to modern technology, both of which slowed the dissemination of advances across sectors such as energy and agriculture. Regulatory barriers, ranked at 17.5%, were deemed substantial, owing to complicated and restrictive legislation and opaque regulatory frameworks that create uncertainty for investors. The remaining institutional, market, environmental, and social barriers, whereas less dominant in relative weight, continue to impede Morocco's green transformation in terms of efficiency and inclusiveness.

The Fuzzy VIKOR methodology offered further clarity on the most efficacious techniques for overcoming these obstacles. The findings indicated that reskilling initiatives for green

TABLE 6 | S, R, and Q values of GTA strategic solutions.

Code	Strategic solutions	S_i	R_i	Q_i
SO7	Resilience programs for green tech jobs	0.2329	0.2267	0.2146
SO6	Banks consolidating digital tools for green investments	0.5366	0.2949	0.2446
SO4	Platform connecting vendors and financial institutions	0.8843	0.4797	0.2972
SO2	Simplifying and modernizing regulatory processes	0.8999	0.8532	0.3532
SO10	Coordinated efforts between key institutions	0.2248	0.763	0.3961
SO3	Demand-side incentives (rebates, tax credits)	0.1565	0.4961	0.47
SO9	Feed-in tariffs (FiTs) with long-term contracts	0.4282	0.3064	0.4815
SO8	Transparent governance and green public procurement	0.804	0.5099	0.5088
SO5	Dedicated green technology R&D funds	0.728	0.5658	0.5657
SO11	Capital subsidies to reduce initial costs	0.1521	0.7909	0.7657
SO1	Green industrial policies for targeted support	0.774	0.7918	0.8239

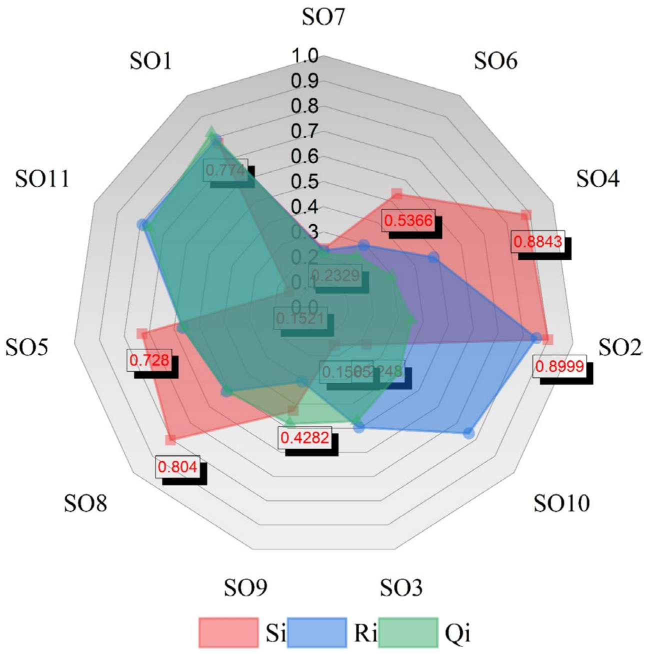

FIGURE 15 | GTA Strategic alternative results based on S_i, R_i, and Q_i.

TABLE 7 | Prioritization of green technology policies according to lowest Q values.

Code	Policies	Q_i	Rank
SO1	Green industrial policies for targeted support	0.8239	11
SO2	Simplifying and modernizing regulatory processes	0.3532	4
SO3	Demand-side incentives (rebates, tax credits)	0.470	6
SO4	Platform connecting vendors and financial institutions	0.2972	3
SO5	Dedicated green technology R&D funds	0.5657	9
SO6	Banks consolidating digital tools for green investments	0.2446	2
SO7	Reskilling programs for green tech jobs	0.2146	1
SO8	Transparent governance and green public procurement	0.5088	8
SO9	Feed-in Tariffs (FiTs) with long-term contracts	0.4815	7
SO10	Coordinated efforts between key institutions	0.3961	5
SO11	Capital subsidies to reduce initial costs	0.7657	10

technology employment (SO7) provide a notably advantageous option, underscoring the significance of human capital advancement in enabling a green transition. Workforce training is essential for equipping Morocco to meet the requirements of emerging green sectors and ensuring that technological adoption is not impeded by skill deficiencies. The second-ranked option is the consolidation of digital instruments for green investments by banks (SO6), highlighting the significance of financial innovation. Improving banks' capacity to evaluate, administer, and allocate capital for green initiatives can significantly reduce financing deficits, particularly for small and medium-sized enterprises. The third-ranked option, platforms that link green technology suppliers with financial institutions (SO4), emphasizes the necessity for enhanced collaboration between innovators and bankers to align funding options more accurately with technological supply and demand. Additional criteria, like the simplification of regulatory processes (SO2) and the enhancement of institutional coordination (SO10), received high scores, underscoring the significance of creating a conducive institutional and policy framework.

The results of this research indicate that Morocco's strategy for expediting the adoption of green technology involves tackling the interactions among finance, technology, and legislation. A limited emphasis on infrastructure or subsidies will

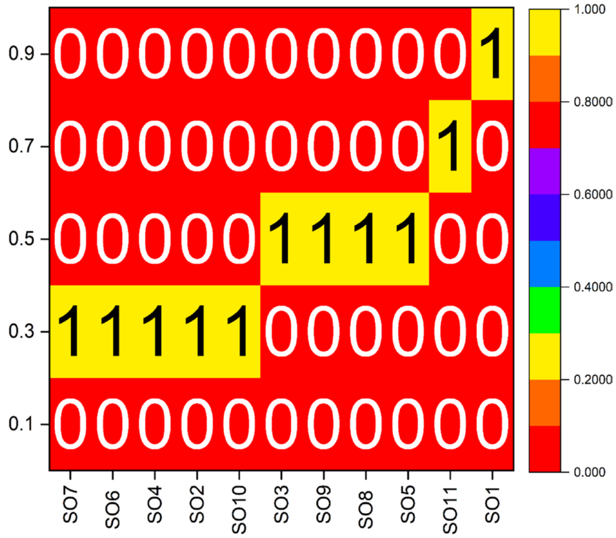

FIGURE 16 | Lowest Q_i-based prioritization of solutions to GTA barriers.

not suffice. Instead, a holistic approach is crucial, combining financial innovation, workforce development, and regulatory reform with wider institutional cooperation. When combined with regulatory modernization and institutional cooperation, these policies provide a viable road to a low-carbon economy. Morocco can achieve its sustainability goals while simultaneously creating excellent jobs, attracting international investment, and positioning itself as a regional leader of green innovation and resilience.

5.1 | Research Implications

This study enhances the conceptual discourse on green technology uptake by integrating actual data from Morocco with established frameworks on systemic innovation and transition management. The Fuzzy AHP method offered a systematic approach to identify, quantify, and prioritize the primary obstacles to adoption, particularly in emerging economies. The approach enhances comprehension of the interactions among financial, technological, and regulatory systems that either facilitate or impede sustainable transitions. It underscores the significance of connecting innovation systems to the reality of local governance and market settings, providing a solid foundation for comparison with nations experiencing analogous socio-economic circumstances. This study enhances theoretical understanding of how institutional capability and economic maturity affect global green transition dynamics by analyzing Morocco in the context of developing economies.

The study outcomes have crucial managerial policy implications for Morocco's efforts to enhance green technologies and overcome its barriers. These findings underscore the pressing necessity for specialized financial instruments such as green bonds, concessional loans, and public-private partnership frameworks to address the financing deficit for practitioners and policymakers. Technology managers should focus on innovation that happens in their own country by putting money into research and development and training programs for workers. This will make them less reliant on solutions that come from other countries. To get more private sector involvement, regulatory bodies need to make bureaucratic processes easier and policies more open. To make sure that green technology projects are good for the environment and for everyone, it will be important to improve collaboration between different sectors and community involvement measures. The suggested alternatives give us a way to build a stronger and more resilient energy infrastructure while also reducing the negative effects of global warming.

A thorough and coordinated plan is needed to successfully get over the barriers to using green technology in Morocco. This should include making the rules easier to follow and more encouraging of private investment as well as upgrading infrastructure to make it easier for green technologies to be used. A critical element of this strategy involves implementing reskilling programs for green technology jobs to build a workforce capable of supporting emerging industries. Financial reforms are equally crucial; on one hand, banks must consolidate digital tools to enhance access to green investments and on the other hand, platforms connecting vendors with financial institutions should be established to facilitate better market integration. Regulatory processes must be simplified and modernized to reduce bureaucratic hurdles and attract private sector participation. Additionally, coordinated efforts among key institutions are needed to harmonize policies, investments, and innovation initiatives. Raising public awareness about environmental protection and sustainability practices is necessary to drive cultural change and foster societal acceptance. Finally, by actively involving local communities and stakeholders in decision-making processes, Morocco can enhance trust, ensure project relevance, and build a participatory framework for sustainable development. These measures will not only address immediate barriers but also lay the foundation for a resilient, inclusive, and thriving green economy.

This study demonstrates the efficacy of integrating fuzzy multicriteria methods in sustainability research. It provides a template for how to systematically link barrier analysis with solution ranking, contributing to the literature on decision-support frameworks for sustainable development. The application of the integrated FAHP–FVIKOR hybrid approach can be adopted to other complex decision problems, suggesting a versatile tool for interdisciplinary studies. By highlighting these implications, this study addresses an immediate case of developing economies and also informs broader policy strategies and methodological approaches in the field of GTA.

Despite the effectiveness of the methodology, this study has certain limitations. This study focused on a Moroccan context with an expert panel of eight, which, adequate for our methodology, may limit the generalizability of the findings. Future research should address these limitations through expanded and comparative analyses. Moreover, a longitudinal study could track how the significance of these barriers evolves over time as new policies and technologies emerge, providing insights into the dynamics of GTA. Similarly, applying our integrated FAHP–FVIKOR framework in other developing economies, such as comparative analysis between different regions, would test the robustness and universality of the barrier rankings and solution effectiveness in different cultural and economic settings.

Additionally, future work might involve a larger expert pool or a Delphi-based approach to refine the weightings and consensus on barriers. By pursuing these directions, subsequent research can build on our findings, address current limitations, and ultimately enhance the strategies for overcoming GTA barriers worldwide.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1:** sd70571-sup-0001-Supinfo.pdf.